Open Salad Mix – Assembly Guide

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Part codes as in the BOM: P = 3D-printed part · M = mechanical part · E = electrical part.

1. Assembly guide

1.1 Product overview and assembly units

The machine consists of 3 units, assembled in the order **Base → Lower roller unit → Upper roller unit**. The numbers in the circles are the part numbers, with the BOM code.

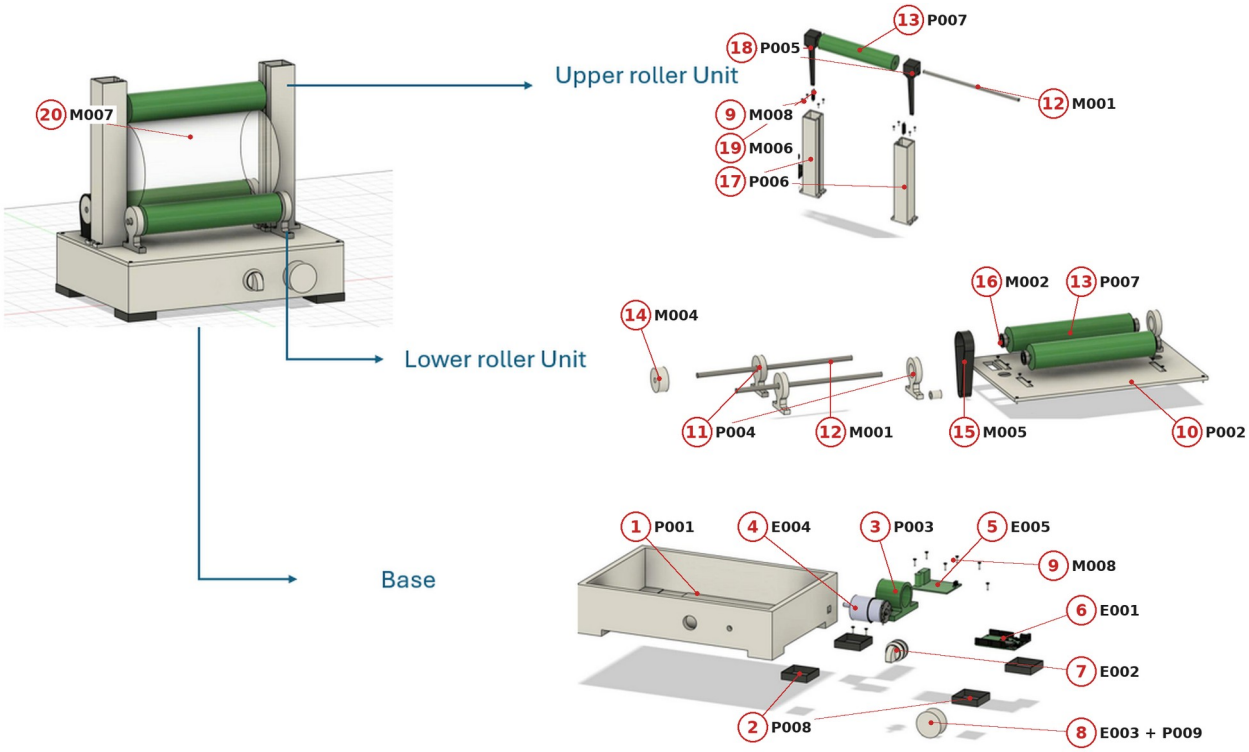


Figure 1 – Open Salad Mix and its 3 assembly units: Base, Lower roller unit, Upper roller unit

No.	BOM code	Part name (as in BOM)	Qty
Base			
1	P001	Base Housing	1
2	P008	TPU Feet	4
3	P003	Motor Holder	1
4	E004	Motor JGB37-520	1
5	E005	Motor Controller (BTS7960)	1
6	E001	Controller (Arduino Leonardo)	1
7	E002	Rotary Selector (knob included)	1
8	E003 + P009	Rotary Timer Switch + Control Knob (timer)	1 + 1
9	M008	Screw 2.5 × 12 mm	30
–	M003	GT2 Pulley 20T (on the motor shaft, hidden in the figure)	1
–	E006, E007	Power Supply 12 V, Electrical Wire (not shown)	1, 1
–	E009	Resistor 10 kΩ – pull-up / pull-down for the controller inputs (not shown)	2
Lower roller unit			
10	P002	Deck Plate	1

No.	BOM code	Part name (as in BOM)	Qty
11	P004	Lower Bearing Holder	4
12	M001	Stainless Shaft – 240 mm, 255 mm (lower rollers); 260 mm (upper roller)	3
13	P007	Roller Sleeve	3
14	M004	GT2 Pulley 60T	1
15	M005	GT2 Endless Belt	1
16	M002	Bearing 608-2RS (4 lower + 2 upper)	6
Upper roller unit			
17	P006	Spring Column / Slider Holder	2
18	P005	Upper Bearing Holder (long arm)	2
19	M006	Extension Spring	2
12, 13, 16, 9		Shared: 260 mm shaft, Roller Sleeve, Bearing, Screw	
Use			
20	M007	PET Jar	1

1.2 Before you start

Tools: screwdriver (to fit the 2.5 × 12 mm screws) · small hex key for the pulley set screws · needle-nose pliers / small hook for the springs · deburring knife · cutter or hacksaw + file for the Ø8 stainless shafts.

Screws: only one screw type is used – Screw 2.5 × 12 mm, coarse thread (M008), driven directly into the printed plastic. **Tighten by hand, do not overtighten** – coarse-thread screws strip PETG easily.

Prepare the parts:

- 3D-print all parts P001–P009 (PETG / PETG-CF; P007 and P008 in TPU 95A). The speed knob comes with the Rotary Selector (E002) and is not printed.
- Cut the 3 Stainless Shafts Ø8 (M001) and deburr both ends: **240 mm** (idle lower roller), **255 mm** (driven lower roller), **260 mm** (upper roller). Mark the length on each shaft so they are not mixed up.

1.3 Assemble the Base

Step 1 – Fit the feet

Parts needed	Assembly sequence
 **TPU Feet** P008 · Qty 4 **Base Housing** P001 · Qty 1	**1.** Turn the Base Housing (P001) upside down, bottom facing up. **2.** Press the 4 TPU Feet (P008) into the 4 corners of the bottom – press-fit, no screws. **3.** Turn the base back over, open side facing up.

Step 2 – Mount the motor in the motor holder

Parts needed	Assembly sequence
 **Motor JGB37-520** E004 · Qty 1  **Motor Holder** P003 · Qty 1  **GT2 Pulley 20T** M003 · Qty 1  **Deck Plate** P002 · Qty 1	**1.** Insert the Motor (E004) into the round opening of the Motor Holder (P003), motor wires pointing towards the controller. **2.** Fit the GT2 Pulley 20T (M003) on the motor shaft, with the set screw on the flat of the D-shaft. **3.** Do not fully tighten the set screw yet – the axial position of the 20T pulley is aligned with the 60T pulley in step 8. **4.** Place the Deck Plate (P002) on the base (no screws).

Step 3 – Screw the motor holder to the base

Parts needed	Assembly sequence
• Motor assembly (step 2)  **Base Housing** P001 · Qty 1  **Screw 2.5 × 12 mm** M008 · Qty 4	**1.** Place the motor assembly in its position in the base (dry compartment), motor shaft facing the belt side. **2.** Screw the Motor Holder to the bottom of the base with 4 M008 screws.

Step 4 – Install the electronics in the base

Parts needed	Assembly sequence
 **Motor Controller (BTS7960)** E005 · Qty 1  **Controller (Arduino Leonardo)** E001 · Qty 1  **Rotary Selector** E002 · Qty 1  **Rotary Timer Switch** E003 · Qty 1  **Control Knob (timer)** P009 · Qty 1  **Power Supply 12 V** E006 · Qty 1  **Electrical Wire** E007 · Qty 1  **Resistor 10 kΩ** E009 · Qty 2  **Screw 2.5 × 12 mm** M008 · Qty 6	**1.** Screw the Motor Controller BTS7960 (E005 – board next to the Motor Holder) to the base with 4 M008 screws. **2.** Screw the Controller – Arduino Leonardo (E001) to the base with 2 M008 screws. **3.** Fit the Rotary Selector (E002) and the Rotary Timer Switch (E003) into the 2 holes in the front of the base and tighten the front nuts; the E002 knob is supplied, only fit the Control Knob (P009) on the timer. **4.** Route the cable of the Power Supply (E006) through the side opening of the Base Housing and connect the wires (E007) according to the wiring diagram. **5.** Connect the 2 Resistors 10 kΩ (E009) as pull-up / pull-down resistors at the controller inputs, as shown in the wiring diagram.

1.4 Assemble the Lower roller unit

Step 5 – Prepare the 2 lower rollers

Parts needed	Assembly sequence
 **Stainless Shaft** M001 · Qty 2 · 240 mm + 255 mm  **Roller Sleeve** P007 · Qty 2  **Bearing 608-2RS** M002 · Qty 4  **Lower Bearing Holder** P004 · Qty 4	**1.** Press the 240 mm and 255 mm Stainless Shafts into 2 Roller Sleeves (P007) → **assembly A** (2 lower rollers). **2.** Press 4 Bearings (M002) into the 4 Lower Bearing Holders (P004) → **assembly B**. **3.** Fit both shaft ends of assembly A into assembly B (2 Lower Bearing Holders per roller).

Step 6 – Mount the 2 lower rollers on the deck

Parts needed	Assembly sequence
• Lower roller assembly (step 5)  **Deck Plate** P002 · Qty 1  **Screw 2.5 × 12 mm** M008 · Qty 8	**1.** Place the 4 Lower Bearing Holders on the Deck Plate (P002) and screw each one down with 2 M008 screws (8 screws in total). **2.** Turn each roller by hand: it must turn smoothly without binding.

Step 7 – Fit the 60T pulley and the belt

Parts needed	Assembly sequence
 **GT2 Pulley 60T** M004 · Qty 1 **GT2 Endless Belt** M005 · Qty 1	**1.** Fit the GT2 Pulley 60T (M004) on the end of the driven roller shaft (belt side) and tighten the set screw. **2.** Feed the GT2 Endless Belt (M005) down through the opening in the Deck Plate; hook the lower part onto the 20T pulley and the upper part onto the 60T pulley. **3.** If the belt is too tight: remove the Lower Bearing Holder, adjust the shaft-end position, then screw the Lower Bearing Holder back on.

Step 8 – Tension the belt and screw the deck to the base

Parts needed	Assembly sequence
• Deck assembly (steps 6–7) • Base assembly (steps 1–4)  **Screw 2.5 × 12 mm** M008 · Qty 4	**1. Align the 2 pulleys:** looking along the belt, slide the 20T pulley along the motor shaft until the belt runs straight without being pulled sideways, then tighten the set screw. **2.** Turn the motor to check that the belt is tensioned. **3.** Screw the Deck Plate to the base with 4 M008 screws at the 4 corners.

1.5 Assemble the Upper roller unit

Step 9 – Press the bearings into the Upper Bearing Holders

Parts needed	Assembly sequence
 **Bearing 608-2RS** M002 · Qty 2 **Upper Bearing Holder (long arm)** P005 · Qty 2	**1.** Press 1 Bearing (M002) into the square housing of each Upper Bearing Holder (P005). **2.** Press straight with even force (vice or flat plate); never hit the inner ring of the bearing.

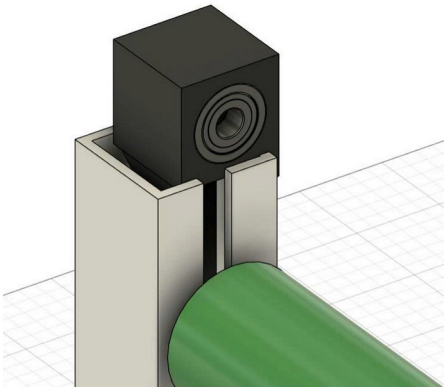
Step 10 – Hook the springs and fit the upper roller

Parts needed	Assembly sequence
 **Extension Spring** M006 · Qty 2 • 2 Upper Bearing Holders (step 9)  **Stainless Shaft** M001 · Qty 1 · 260 mm **Roller Sleeve** P007 · Qty 1	**1.** Hook the **lower end** of the Extension Spring (M006) into the anchor point on the Deck Plate. **2.** Hook the **upper end** of the spring into the hole at the end of the arm of the Upper Bearing Holder (P005). **3.** Repeat on the other side. **4.** Press the 260 mm Stainless Shaft into a Roller Sleeve (P007). **5.** Fit both shaft ends into the bearings of the 2 Upper Bearing Holders.

Step 11 – Mount the 2 Slider Holders on the deck

Parts needed	Assembly sequence
 **Spring Column / Slider Holder** P006 · Qty 2 **Screw 2.5 × 12 mm** M008 · Qty 8	**1.** Hold the Roller Sleeve and pull the Upper Bearing Holders up to a vertical position. **2.** With the Upper Bearing Holder vertical, slide the Slider Holder (P006) on from above (groove of the Slider Holder facing inwards). **3.** Repeat on the other side. **4.** Screw each Slider Holder to the Deck Plate with 4 M008 screws (8 screws in total).

Step 12 – Place the jar and check the open position

Parts needed	Assembly sequence
 **PET Jar** M007 · Qty 1	**1.** Lift the upper roller to the end of its travel and rest **the notch at the bottom of the square housing of the Upper Bearing Holder on the top edge of the Slider Holder** (open position) on both sides. **2.** Lay the PET Jar (M007, lid tightly closed) horizontally on the 2 lower rollers, centred along the rollers. **3.** Lift the Upper Bearing Holders off the Slider Holder edge and lower the upper roller slowly until it rests on the jar – do not let it drop.  *Open position: notch at the bottom of the Upper Bearing Holder rests on the top edge of the Slider Holder*

2. Checks after assembly

Check in this order: **unplugged → plugged in, without jar → with jar**. If a check fails, stop and fix it before continuing.

#	Check	Pass when
	A. Unplugged	
A1	All screws fitted, no loose screws inside the machine	All 30 M008 screws used: base 10 (Motor Holder 4, Motor Controller 4, Controller 2) · deck 12 (Lower Bearing Holders 8, deck to base 4) · 2 Slider Holders 8
A2	Turn the driven roller by hand	Both lower rollers and the motor turn smoothly; the 2 pulleys are aligned, the belt runs centred on the pulleys without drifting, skipping teeth or rubbing
A3	Lift the upper roller to the open position and rest it on the Slider Holder edges	Held on both sides, does not drop by itself
	B. Plugged in, without jar	
B1	Plug in, Rotary Timer Switch at 0	Motor does not run
B2	Rotary Selector at position 1, set the timer	Motor runs slowly and stops when the time is up
B3	Change the Rotary Selector position	Motor speed changes
	C. With jar	
C1	Place the PET Jar filled with water to 55–65 %, lid closed, lower the upper roller	Jar sits centred, upper roller rests evenly
C2	Run 30 s at each speed position	Jar turns evenly, no slipping, no axial creep, no leaks